

Auditing Recommender Systems for User Empowerment in Very Large Online Platforms under the Digital Services Act

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Abstract

The governance of recommender systems (RSs) in very large online platforms (VLOPs) is expected to change significantly under the Digital Services Act (DSA), which imposes new obligations on transparency and user control. However, beyond legal compliance, a critical question remains: *How can recommender systems be redesigned to genuinely empower users and foster meaningful personalization?* This paper addresses this question by analyzing how three major short-video platforms—Instagram, TikTok, and YouTube—have implemented the DSA requirements for RSs. By reviewing their audit reports, systemic risk assessments, and compliance strategies, we evaluate the extent to which current approaches enhance user autonomy and control over content exposure. Building on this analysis, we outline a perspective for the future of VLOPs' RSs grounded in speculative design. We argue that meaningful personalization should integrate algorithmic choice, balancing proportionality and granularity in RS customization, and content curation, ensuring diversity and authoritativeness to mitigate systemic risks. By bridging legal analysis, platform governance, and user-centered design, this paper outlines actionable pathways for aligning technical developments with regulatory objectives. Our findings contribute to interdisciplinary research on RSs by highlighting how platforms can move beyond minimal compliance toward a model that prioritizes user empowerment and content pluralism.

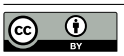
CCS Concepts

• **Social and professional topics** → **Computing / technology policy**; • **Information systems** → **Recommender systems**; **Personalization**; • **Human-centered computing** → *Interaction design.*

Keywords

Recommender Systems, Algorithmic Auditing, User Empowerment, Very Large Online Platforms, Digital Services Act

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1 Introduction

In the contemporary digital age, recommender systems (RSs) shape the way in which people interact online and offline: from social media to music streaming, from e-commerce to news websites, suggested content and products get the spotlight on platforms' interfaces and influence individuals' interests and priorities. Despite their usefulness in reducing information overload, the nudging potential of RSs may yield harmful effects on users [40], spanning from the amplification of disinformation to an unaccountable shaping of individual perspectives and public opinion. The governance of RSs, especially those deployed by very large online platforms (VLOPs) and search engines (VLOSEs), has drawn considerable regulatory attention due to their potential to impact electoral processes, users' autonomy and physical and mental well-being. In this context, the European Union's Digital Services Act (DSA) [1] has introduced the first supranational regulatory framework aimed at enhancing transparency and empowering users with meaningful control over algorithmic recommendations. However, whether the measures outlined in the DSA are sufficient to address the challenges posed by RSs remains an open question. This paper situates the problem of user empowerment in social media platforms within the emerging debate on the regulatory aspects of RSs, which influence prominent real-world applications of this technology.

A policy focus has emerged regarding the potential risks of RSs, such as the negative impacts on minors of the "rabbit hole effect" [17], the influence on voting intentions [9], and the challenge of assessing the truthfulness of conflicting news [61]. However, there is no conclusive evidence that RSs, as an element of platform design, directly cause these effects [39, 51]. This uncertainty raises several challenges about how legal obligations should be designed to ensure meaningful user control, balancing algorithmic personalization with the need for information diversity, and about the extent to which platforms' implementation of relevant regulatory requirements aligns with societal expectations. This research explores the interplay between the DSA, its application by platforms, and the design of RSs by addressing the following research questions:

> **RQ1**. *What do the EU legal requirements for RS transparency and control entail, and how have they been implemented by selected VLOPs?*

> **RQ2**. *Mediating between regulatory principles and platforms' practices, how can users be empowered, during their interaction with a RS, to achieve a meaningful personalization of their online experience?*

In this study, we aim to provide a perspective on how legal, technical, and ethical dimensions may converge to shape the future of personalization in online platforms. Firstly, we analyze the

transparency and user control obligations introduced by the DSA and their connection with related EU regulations (Section 2). This discussion establishes the regulatory foundation for RS governance. Secondly, we examine the compliance strategies of three social media VLOPs—TikTok, Instagram, and YouTube—by reviewing their DSA audit and systemic risk assessment reports. This analysis highlights key gaps between regulatory expectations and platform practices, presented in Section 3. Thirdly, we propose a speculative design framework for RS governance in Section 4, outlining design features that could foster meaningful personalization while enhancing users' autonomy and control over recommendations. Finally, we integrate insights from legal, technical, and ethical considerations to outline actionable pathways for aligning regulatory goals with technical affordances and discuss future directions in Section 5.

2 Regulatory background

The DSA is the first supranational regulation addressing the transparency and controllability of RSs with the aim of empowering users of online platforms [24, 52]. In particular, art. 27(1) requires platform providers to explain "in their terms and conditions, in plain and intelligible language, the main parameters used in their recommender systems, as well as any options for the recipients of the service to modify or influence those main parameters". The rationale of this provision is to "ensure that recipients of their service are appropriately informed about how recommender systems impact the way information is displayed, and can influence how information is presented to them" (recital 70). Therefore, the parameters considered must include, at least, "the criteria which are most significant in determining the information suggested to the recipient of the service" (content) and the reasons for its "relative importance" (ranking) (art. 27 (2)). Additionally, when options to modify or influence the main parameters are mentioned in the terms and conditions, platforms should provide, in correspondence with the list of ranked recommendations, a "directly and easily accessible" functionality "that allows the recipient of the service to select and to modify at any time their preferred option" (art. 27 (3)). Moreover, according to art. 38, VLOPs that use RSs "shall provide at least one option for each of their recommender systems which is not based on profiling". The aim of these provisions is to foster users' empowerment through their awareness of how the recommendation process works and their direct intervention on the platform's interface. The implementation of these articles is an ongoing process whose results differ across VLOPs, while, regarding most online platforms that are not VLOPs, compliance has not been assessed yet. It should be noted that the design of RSs can enable the emergence of systemic risks within the platform environment (art. 34(2)), impacting, among others, the exercise of fundamental rights, civic discourse, and electoral processes, public security, the protection of minors, and people's physical and mental wellbeing. VLOPs and VLOSEs should identify (art. 34(1)) and mitigate (art. 35) systemic risks through interventions to be documented in publicly accessible reports, which represent a source for scrutinizing the implementation of art. 27 and 38, together with audits.

Observation 1. *VLOPs' compliance with the DSA provisions on RSs is inconsistent due to the lack of technical definitions and implementation standards by regulators.*

Definitions of key terms. Some audit reports provide hints on how the terms of art. 27 have been interpreted by VLOPs. According to Instagram's DSA audit [22], redacted by EY, main parameters should be intended as "the top Predictive Events and their corresponding top Signals, with a Predictive Event being the output of a machine learning model that predicts a probability that some event or action will occur, and Signals being the raw, feature-level inputs that feed into the predictive model and determine what the predictive event will be for each user". The same document describes the most significant criteria as "top predictive events and their corresponding top signals", thereby repeating the definition of main parameters. Given the regulatory uncertainty in this regard, we propose a tentative breakdown of the terminology of art. 27:

Definition 1. `Parameters.` *The signals used by the system to select and rank candidate recommendations (e.g., watch time).*

Definition 2. `Criteria.` *The metrics underlying how certain information or parameters are prioritized in the recommendation process (e.g., the relevance of watch time in selecting and ranking candidate recommendations).*

Definition 3. `Options.` *The available control features provided to users to influence or modify the parameters (e.g., the possibility to select different levels of content sensitivity).*

Definition 4. `Functionalities.` *The specific interface elements that allow users to choose an option to influence the system's parameters (e.g., selectors for levels of content sensitivity).*

Regulatory intersections. A promising development in this regard is the approval of the Code of Conduct on Disinformation (CCD) by the Commission and the Board of Digital Services on 13 February 2025. The CCD is the result of the integration of the 2022 Strengthened Code of Practice on Disinformation within the DSA compliance framework. As a result of this process, the CCD "will become a relevant benchmark for determining DSA compliance regarding disinformation risks for the providers of VLOPs and VLOSEs that adhere to and comply with its commitments" [19].

Five VLOPs (Facebook, Instagram, TikTok, YouTube, LinkedIn) and two VLOSEs (Bing, Google Search) are among the signatories of the Code: hence, for these providers, applying the CCD commitments will be taken as evidence of compliance with the related DSA provisions. This is highlighted by the Commission's opinion on the CCD, according to which "Certain Commitments (such as for example Commitment 19 on transparency of recommender systems) are drafted in a language that mirrors that of certain provisions of Regulation (EU) 2022/2065. When this is the case, the Code of Practice on Disinformation represents a further operational and complementary layer, including reporting obligations, on top of existing legal obligations" [20]. Commitment 19 requires "Signatories using recommender systems [...] to make them transparent to the recipients regarding the main criteria and parameters used for prioritising or deprioritising information, and provide options to users about recommender systems, and make available information on those options". This Commitment explicitly requires signatories to provide options for users to control RSs and, as the first benchmark for compliance with the DSA provisions on transparency and controllability of RSs, offers a grounded perspective on how to interpret the requirements of art. 27: VLOPs that signed the CCD

will need to provide control options for RSs beyond the possibility of disabling profiling-based recommendations as per art. 38.

The CCD also requires signatories to "give users the option of having signals relating to the trustworthiness of media sources into the recommender systems" (Measure 22.2) and publish "information outlining the main parameters their recommender systems employ in this regard" (Measure 22.3). These requirements aim to detail the user control provisions for RSs in the direction of the trustworthiness of recommended content. The responsibility of defining and incorporating trustworthiness signals into platforms' RSs might set new scenarios for the online information ecosystem: in practice, major platforms would have to publicly state their trustworthiness indicators for content and how these are used as signals to feed RSs. No VLOP has apparently implemented these provisions as of now.

The European Media Freedom Act (EMFA) [2] advances relevant requirements for the source of recommended content. While not directly related to RSs, in fact, its art. 3 introduces the "right of recipients of media services to have access to a plurality of editorially independent media content", also in online platforms. According to its art. 18, media service providers can self-declare to VLOPs, based on criteria of editorial responsibility and independence, to enjoy additional safeguards against content moderation practices such as visibility restrictions and demotion through RSs. Moreover, the European Board for Media Services "shall regularly organise a structured dialogue between providers of very large online platforms, representatives of media service providers and representatives of civil society" in order to "foster access to diverse offerings of independent media on very large online platforms" (art. 19(1) EMFA).

These requirements outline principles of diversity and authoritativeness for the provision of media content in VLOPs, complementing the user empowerment and trustworthiness perspectives emerging from the DSA and CCD. These principles, considered together, can substantiate meaningful personalization in the access to information, by increasingly giving users control over what they can, and therefore may want to, see online. The achievement of meaningful personalization in the context of the DSA depends, however, not only on the existence of legal provisions, but also on their interpretation and application. While the DSA enforcement process lacks transparency and therefore accountability due to the almost complete unavailability of relevant legal documents [26], much of the leverage for a substantive implementation of the regulatory requirements on RSs is currently left to the benchmarking process developing between VLOPs, independent auditors, and the Commission. The next section examines the state-of-the-art of the implementation of these requirements in three social media VLOPs.

Observation 2. *The CCD introduces a compliance benchmark for user control and trustworthiness of RSs under the DSA, while the EMFA supports diversity and authoritativeness of recommended media content.*

3 Analysis of the implementation

Building on the previous legal analysis of the EU regulation supporting the transparency and user control of RSs, we now examine its implementation by three social media VLOPs mainly focused on short-video content: Instagram, TikTok, and YouTube. These platforms have been chosen for two main reasons. Firstly, these are among the most used platforms for content discovery by young users and, through their short-video format, reinforce a flow experience that may stimulate addictive behavior [42, 49], even if this has been contested [39]. Secondly, and relatedly, these have been a target of the European Commission investigations under the DSA concerning RSs, especially regarding their impact on mental well-being and political influence.

Instagram is undergoing infringement proceedings for the design of its interface, "which may exploit the weaknesses and inexperience of minors and cause addictive behaviour, and/or reinforce so-called 'rabbit hole' effect" [10]. TikTok is subject to three proceedings, which concern, on the one side, its "persuasive design patterns", considered "problematic for the physical and mental well-being" of users [11], and, on the other side, the role of its RSs in facilitating "coordinated inauthentic manipulation or automated exploitation of the service", supposedly linked to the recent Romanian presidential elections case [9]. Finally, YouTube, while not currently undergoing proceedings, has received a request of information from the Commission in October 2024 regarding the "parameters used by" its RSs, "as well as their role in amplifying certain systemic risks, including those related to the electoral process and civic discourse, users' mental well-being (e.g., addictive behaviour and content 'rabbit holes'), and the protection of minors" [18]. The compliance with the provisions on RSs is therefore one of the main areas in which these VLOPs are being held accountable under the DSA.

We carried out an analysis of the implementation of the DSA requirements on RSs based on three DSA-related documental sources for each of the selected VLOPs: the audit report, the audit implementation report, and the systemic risk assessment report. The independent auditors, EY for Instagram and YouTube and KPMG for TikTok, mostly relied on a walkthrough approach [38] to highlight potential mismatches between the documentation provided by VLOPs and the actual platform design. After reviewing the parts of the aforementioned documents referring specifically to art. 27 and 38 DSA, we integrated the analysis of auditors' findings and platforms' statements with the additional documentation available in the VLOPs' Transparency Centers, to which some reports refer. Based on the evidence collected through this review, we performed an ad-hoc walkthrough on the mobile interface of each VLOP to verify whether the publicly available information provided in compliance with the DSA corresponds to the options and functionalities accessible through the platform's interface. Due to space limitations, our analysis cannot granularly consider all relevant aspects of the aforementioned reports, but it will highlight the information that is fundamental to assess VLOPs' compliance with the DSA provisions on RSs (art. 27 and 38) and their implications for systemic risks.

3.1 Instagram

Audit report. The outcomes of Instagram's audit [22] are positive for all the paragraphs of art. 27 and 38. The audit mostly relies on Meta's system cards [47], which describe Facebook and Instagram RSs parameters and control options. In the case of Instagram, ten RSs are described (Feed, Stories, Explore, Reels chaining, Search, Feed recommendations, Suggested accounts, Notifications, Comments, Threads Feed), along with the signals influencing each of them and the related control options for users. The reported signals

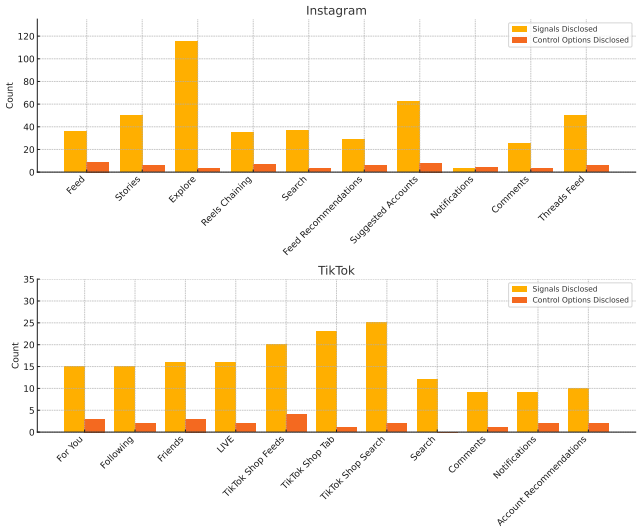


Figure 1: Comparison of signals and control options disclosed in Instagram's and TikTok's audit reports, broken down by recommender system type.

are 442, while the control options 55. Figure 1 shows the breakdown of signals and controls for different RSs types. Where "relative importance" as per art. 27(2) is intended to mean the relative weight of single parameters in influencing the recommendations, the cards do not explain the relative importance of parameters, nor the reasons for their relative importance: therefore, under this interpretation, Instagram would not be compliant with art. 27(2b), although the audit reasonably assures otherwise, maybe based on a different unstated interpretation. Moreover, not all the control options available are described: indeed, in the control dashboard for reels, there is an option enabling users to choose which amount of political content from people they do not follow they want to see (more, standard, less), but this option is not stated in the relevant system card (Feed recommendations > Manage suggested content). Given the positive assessment of all the aspects related to art. 27 and 38, Instagram's audit implementation report [46] does not address these.

Aside from compliance considerations, it can be observed that the signals disclosed are both implicit (e.g., watch time) and explicit (e.g., like), with a prevalence of the former. The control options, apart from those allowing the disabling of personalization as per art. 38, which result in a ranking of posts based on reverse chronological order, are mostly related to traditional user interaction features, such as *save, mute/unmute, follow/unfollow, hide, report*. We argue that these options do not allow users to intervene directly on which category of content is prioritized, thereby preventing them from effectively steering recommendations towards their preferences. The options aimed at providing more direct controllability are available just for content recommended from accounts that a user does not follow and include: *interested/not interested, specific words and phrases*, to hide recommended content containing such expressions in hashtags or captions, *change sensitive content* and the above mentioned *change political content*, both allowing users to choose which amount of content classified as "sensitive" or "political" they will be exposed to (more, standard, less are the options). The rationale for setting control options with different granularity according to the specific RS is not explained: in practice, only Feed recommendations can be controlled through more sophisticated approaches (e.g., *change sensitive content*), even though the impact of selecting the related options on the underlying algorithm is not explained.

> **Observation 3**. *The detailed lists of parameters and options in Instagram's RSs cards do not translate to granular explanations and user control.*

Systemic risk assessment report. As regards Instagram's systemic risk report [48], Meta firstly highlights that its RSs "are designed to try to prevent the recommendation, recirculation, or amplification of potentially policy-violating or otherwise problematic content". The VLOP actively practices demotion as a means of policy enforcement: "using our repository of signals, our algorithms may down-rank content for a variety of reasons and do not recommend certain accounts, or content, such as those related to repeat policy violators in feeds". Meta has identified four domains of intervention that would help identify and address the systemic risks enabled by its RSs: "Cross-functional Integration", i.e., "the ability to ingest signals from cross-functional teams at scale to help inform enforcement against violating content in a consistent manner"; "Abuse of Hashtag Feature", i.e., "preventing manipulation of hashtags intended to distribute policy violating content"; "Recommendation Features", as recommendations of accounts to follow "could connect threat actors to minors"; and "Content Ranking", whose risk of leading users to exposure to harmful content is reduced "by limiting the role of shares and comments in the distribution of sensitive topics". Apart from fostering user empowerment through the explanations in the RSs cards, Meta claims to offer "users the ability to control how much of certain types of content (including sensitive or low quality content) they see in their feeds", even if no such thing as a "low quality content control" can be found either in the system cards or on the platform. The report does not provide any details on how the technical features of Instagram's RSs are tied to specific systemic risks and which related interventions Meta has designed.

3.2 TikTok

Audit report. According to its Help Center [56], the TikTok RSs include For You, Search, Notifications, Account recommendations, Following feed, Friends tab, LIVE feed, and TikTok Shop, which recommends commercial products through some of the preceding RSs. Also the STEM feed [59], which prioritises educational content related to scientific subjects, should be added to these, even if it is not mentioned in the documents reviewed. The main parameters to predict which content a user might like are: user interactions (e.g., likes, comments, shares, watch time, and other engagement metrics), content information (e.g., hashtags, captions, and soundtracks), and user information (e.g., profile details and account settings) [59]. Each of these main parameters is explained through the signals relevant to it: for example, for the For You Feed (FYF), the signals coming from user interactions include "content you like, share, comment on, and watch in full or skip, as well as accounts of followers that you follow back" [56]. The options to modify or influence the main parameters primarily enable users to exclude or include specific content from their feed, encompassing, again

according to the Help Center: *not interested, refresh your feed, filter keywords, unfollow, manage notifications*, *filters* and *sort by* (only for Shop Search), *unsync your contacts* and *remove suggested accounts* (only for Account recommendations). The signals disclosed are in total 170, while the control options 22. The breakdown of signals and controls for different RSs types is shown in Figure 1. The assessment of compliance with art. 27(2) is positive in TikTok's audit report, although the same consideration made above for Instagram regarding the weight of parameters applies here. However, the report, drafted by KPMG, negatively evaluates the implementation of art. 27 (1) and (3) and 38 and draws a more substantial interpretation of their requirements [35]. In fact, regarding the negative outcome for both art. 27(1) and 27(3), KPMG "identified that the design of the controls in place and their respective control objectives were not sufficient and appropriate to meet the Specified Requirements". The main issue raised concerns the "options to modify or influence those main parameters" (art. 27(1)), which are not deemed sufficient for the following reasons:

> "1. options must enable users to influence or modify the 'main parameters', as defined by TikTok (user information, content information, interaction information) and therefore affect the information that will be presented to the user; and 2. options must be a tool or a feature which the user engages with, over and above a user interaction (defined as a main parameter and already informing the user in the Help Center article that how they use the app / interact with content, will impact what they see) [...] 3. the following controls are not considered to be 'options to modify or influence those main parameters':
> (1) Limiting pool of content: these are content controls that TikTok offers to users that restrict certain content from being retrieved by the recommender systems but do not 'modify or influence' the main parameters (listed above) of the recommender systems.
> (2) Non-personalized options: these offer users separate experiences that are not based on profiling.
> (3) Settings: these are user preferences that affect the user's experience on the Platform generally (like language and translation preferences and location services)".

These motivations attempt to provide a consistent definition of the scope of the terms used by art. 27 and 38, introducing the concept of option as "a tool or a feature which the user engages with, over and above a user interaction", which is instead one of the main parameters for TikTok. Options should give users a substantial possibility to influence the main parameters and should be made accessible through the functionality of art. 27(3). For example, according to this perspective, an option should not consist in allowing users to simply disable watch history in the settings of their profile, but instead in granting them the right to enable or disable the use of their watch history as a parameter for specific types of RSs at certain intervals and with respect to certain content categories. KPMG considers the non-personalized options provided according to art. 38 as outside the scope of the "options to modify or influence those main parameters": while the non-personalized options are indeed options, their exclusion from the scope of art. 27 according to KPMG might be due to the fact that they are already covered by art. 38 and so not sufficient for substantially complying with the former provision. The negative outcome of the audit for art. 38 amounts to the observation that, for the Stories RS, the non-personalized option was not available during part of the evaluation period: a less prominent problem than the previous one.

Observation 4. *TikTok should provide more substantial control options for its RSs following the auditor's assessment.*

Systemic risk assessment report. For what concerns RSs, the systemic risk report of TikTok [58] focuses in particular on the protection of minors and information integrity in the context of the For You Feed (FYF), adopting measures such as "banning certain hashtags and/or blocking searches associated with emerging risks": as a result, harmful content may be either demoted or unavailable in FYF. TikTok offers users "tools to diversify the content displayed, to understand why videos have been recommended, to choose that certain keywords will not be displayed to them and to reset their FYF as if they were new to TikTok": these explanations and control features are at the center of the platform's efforts to tackle systemic risks emerging through RSs by empowering users to control the dissemination of both theirs and others' content. The connection between these design principles and the technical affordances of the platform is not clarified, and this is reflected by the negative audit result related to the control options described above. In the audit implementation report [57], TikTok does not specify how it will practically address the shortcomings of the control options provided for its RSs from a socio-technical perspective; instead the VLOP declares that "it will complete its assessment of relevant processes and systems to identify any risks to compliance objectives that address the existing options, and, as appropriate, implement new controls or strengthen existing controls over these risks". This sentence is repeated throughout the whole report, and not just for the provisions regarding RSs.

3.3 YouTube

Audit report. The audit report of YouTube, drafted by EY, aligns with that of Instagram [22]: the auditor confirms compliance with all the provisions of art. 27 and 38 [23] and, for this reason, YouTube's audit implementation report does not provide further information in this regard [29]. Unlike Instagram and TikTok, YouTube's audit report does not mention different RSs but focuses on listing a series of high-level parameters and options. The former mostly focus on user interaction data and include: "watch history, search history, channel subscriptions, watch time, sharing, likes/dislikes, video clicks, user selection of "Not interested" and "Don't recommend", survey responses, and Google Activity". The latter are mostly aimed at granting users' control of the data available for feeding recommendations and include:

> "Remove, pause, or delete specific videos from watch history, and/or searches from search history through the Google account, Turn off or delete watch and search history, Ads personalization and additional settings that may impact data available for recommendations such as Choose topics in a user's recommendations on the Home Page and Watch page, Remove recommended content from homepage, Clear "Not interested" and "Don't recommend channel" feedback, Manage recommendations and search results on Google Account activity, Manage liked videos (i.e., remove liked videos on YouTube), Subscribe and Unsubscribe to YouTube channels and Delete survey answers".

While YouTube's parameters are more than 80 billion according to the VLOP's 2024 transparency report under the Strengthened Code of Practice on Disinformation [27], the areas in which they have been grouped are 12 and there is a one-to-one matching between the parameters and the related control options. The options allowing to remove videos from, or outright delete and turn off, watch history, search history, and Google account activity are taken as evidence of compliance with art. 38. Therefore, differently from Instagram and TikTok, YouTube does not have a single specific functionality for disabling profiling-based recommendations, but it allows the user to do that through individual interventions on various parameters. The only control option allowing users to actively choose the content of recommendations is *Choose topics in a user's recommendations on the Home Page and Watch page*, which consists in the selection of topic tags "based on your existing, personalized suggestions" and "on content related to what you interact with", as explained by YouTube Help [63]. This option is complemented by YouTube's search filters, which, apart from providing standard filtering tools (based on chronological order, view count, length of a video) also include tags allowing users to rank search results based on their previous interaction with the video (such as "watched" and "unwatched") [62]. This is the only example of personalized search filter found in the VLOPs considered. The other control options rely on a negative feedback from the user (e.g., *Not interested, Don't recommend channel*).

Systemic risk assessment report. According to YouTube's systemic risk report [28], while "Poorly designed or controlled recommender systems may increase the risk that harmful content goes viral", "properly functioning recommender systems should decrease risk by increasing the visibility of high-quality and trustworthy content and by promoting a diversity of topics and sources for users to explore". This high-level statement hints at YouTube's view on DSA-oriented recommendations, which can mitigate, or at least not enable, systemic risks. In particular, the proper functioning of RSs (which, in this context, may equate to their alignment with regulatory principles) depends on their design and control and results in the promotion of authoritative and diverse content. As can be seen, these principles reflect those emerging from our previous legal analysis: on the one side, user empowerment through control over the design of RSs (as per the DSA and the CCD) and, on the other side, authoritativeness and diversity (as suggested by EMFA). However, the report does not mention how YouTube would operationalize these principles into the functioning of RSs. It does, however, underline that one of the roles of RSs is to support content moderation by "elevating high-quality and trustworthy content", which is considered "a more proportionate approach to addressing harmful content risk than removing content altogether", as this "can present risks to freedom of expression and information".

Observation 5. *YouTube claims to promote authoritative content and foster diversity to address RS-enabled systemic risks.*

3.4 Comparing the emerging evidence

Differing interpretations. The emerging evidence regarding RSs governance in the three VLOPs considered above allows us to draw preliminary considerations on both auditing and compliance practices. First of all, the fact that the auditor of TikTok is different from that of Instagram and YouTube invites us to compare how they respectively inspected the platforms' algorithms. Both EY and KPMG adopted a walkthrough approach to verify whether the parameters, options, and functionalities of RSs described in the platforms' documentation were consistent with what users can find while using their interfaces. It is not stated whether such a walkthrough would be performed on the app or desktop version of the platform, nor which phases it entailed at a granular level. As [38] present it, the walkthrough method consists of "systematically and forensically step[ping] through the various stages of app registration and entry, everyday use and discontinuation of use", and allows to collect "a foundational corpus of data" that can be used to assess an "app's intended purpose, embedded cultural meanings and implied ideal users and uses". In the case at hand, the walkthrough merely focused on the technical design features of the VLOPs, leaving societal and user-centric considerations in the background.

On the legal side, however, EY and KPMG adopted different perspectives. Indeed, while EY positively assessed compliance regarding all the DSA provisions on RSs for Instagram and YouTube, KPMG did the opposite for TikTok, except for compliance with art. 27(2). The main point here concerns the interpretation of which control options should be made available for RSs according to art. 27(1) and (3). EY reasonably assured that the options and functionalities available were sufficient for compliance with art. 27, while KPMG took the opposite stance, considering that options to limit the pool of recommended content, non-personalised options and settings are outside the scope of art. 27. In fact, since the disclosed options for controlling RSs differ only marginally between Instagram, YouTube, and TikTok, their positive or negative evaluation, it can be argued, mainly depends on the interpretation of what substantive control options would entail. In particular, the three platforms have standard options that allow to modify user profile and engagement information, including *unfollow, dislike, not interested* associated with content to be excluded from the recommendation process. What is lacking, with respect to the array of parameters disclosed, are the options allowing users to actively determine their exposure to different types of recommended content: in this perspective, only *change sensitive/political content* on Instagram, the STEM feed and *refresh your feed* on TikTok, followed by the selection of topics in the initial preference elicitation, and Youtube's tags allowing users to explicitly include specific topics in the recommendations go in the direction of empowering users' active choices. However, for what concerns Instagram, the impact of selecting "more", "standard" or "less" for the *change sensitive/political content* options on the inner functioning of every RS is not explained. As regards TikTok, the possibility of only starting from scratch by refreshing the feed clashes with users' interest in keeping some aspects of their feed that would be lost. Lastly, for YouTube, users cannot autonomously input their topics of interest but just select tags within a limited and opaquely generated list.

The emerging market of algorithmic audits. The motivations behind different auditing practices, and the related compliance efforts by platforms under the DSA, have been addressed by [55], who argue that "the breadth of systemic risks that the DSA requires to be considered also allows considerable room for framing, bringing the auditor's priorities and worldview and constructing the DSA requirements around them, rather than the other way around". The

auditors' priorities also include maintaining good business relationships with VLOPs that may already use their services for financial audits, as the Delegated Regulation on the Performance of Audits (DRPA) [16] clarifies that concerns for conflict of interest "should not exclude auditing organisations who have performed statutory financial audits" (recital 8 DRPA). While "responsibility for the initial formulation of the benchmarks against which compliance is/will be sought, rests with the audited organisation (DRPA art. 5 (1) (a))", the "back-and-forth between the auditing organisation and the [...] audited organisation" may "impact on the way auditing benchmarks are formulated and/or standardised in the future" [55]. Consequently, given that, de facto, "the primary responsibility for the formulation of benchmarks is left to the auditing organisation, [...] benchmark disparities amongst different auditors may incentivise platforms to choose their assessors and auditors based on their benchmarks (easy or difficult, simple or complicated)" [55].

Based on these considerations, it can be argued that the market of algorithmic auditing for RSs is going to expand but not necessarily by diversifying its offer: it might indeed align along minimalist interpretations of art. 27, causing the usual concern of transparency washing [64]. Benchmarking disparities regarding which control options are needed to comply with art. 27 may not be solved unless a conscious regulatory intervention, i.e., through guidelines or standards, addresses the terminological and design ambiguities found in the DSA provisions on RSs. Moreover, the partial public disclosure of audit and systemic risk reports has not substantially clarified how platforms (plan to) mitigate the harms emerging from the influence of RSs and whether concrete socio-technical interventions are envisioned to substantiate the user empowerment enshrined in the EU regulatory framework. The next yearly round of DSA-related audits will shed light on whether the risk of minimal compliance persists and more substantial steps towards clear benchmarking have been taken by VLOPs in the meantime.

Observation 6. *VLOPs' diverging interpretations of the DSA provisions on RSs may incentivize minimal compliance.*

4 Speculative design for user empowerment

The prevalent paradigm in RSs is to optimize based on user feedback, without requiring any direct user intervention. However, several studies suggest that users benefit (e.g., through better recommendation acceptance) when they have some degree of control over the system [6, 30, 34, 43, 53]. So, there seems to be some space for designing controllable RSs, where the user may tune the RS's output via some interface. The design of such an interface involves several key decisions, including the selection of appropriate UI elements [12, 41], the granularity of user responses [8], and the most effective way to label user choices [3]. A substantive implementation of art. 27 and 38 DSA [25] would require VLOPs to put in place effective options for users to intervene on the algorithmic parameters to change the output of the recommendations. The complexity of these options should be gradual enough to meet the heterogeneity of skills across users [32]. At the low level, users would be able to give feedback on the recommendations they receive, on its topic and on the content creator (like/dislike) and could see explanations in discursive or graphic form (e.g., word clouds). At the higher level, users could be allowed to modify the degree of personalization of the recommendations, e.g., by choosing the percentage of personalized recommendations they want to see. They should also be able to decide which types of their data, including profile information and interaction signals, can be used as input for RSs.

Existing work on RSs controllability has primarily focused on user interface affordances and transparency mechanisms [12, 44]. However, these efforts often remain constrained by the current platform-centric model, where transparency and user control are treated as auxiliary features rather than fundamental design principles. In contrast to this, we adopt a speculative design approach [4, 5] that integrates proportionality, granularity, diversity, and authoritativeness as core components of RS architecture. The role of speculative design is "first, to enable us to think about the future; second, to critique current practice" [4]: in the context of our work, this approach allows to overcome the aforementioned issues in DSA implementation with actionable principles for meaningful personalization through transparent and controllable RSs. By shifting the focus towards user-driven re-weighting and exposure-aware ranking, our framework envisions a recommendation process where personalization is not just a system-imposed feature but a user-driven mechanism for content curation. Therefore, we propose four design perspectives to enable meaningful personalization and we group them into two broader categories: (1) *enabling algorithmic choice*, which includes proportionality and granularity, and (2) *directing content curation*, which includes diversity and authoritativeness. We also highlight directions for interdisciplinary research bridging regulatory incentives with design innovation in RSs.

Enabling algorithmic choice. This category concerns mechanisms through which users can influence *how* the recommendation process works.

Proportionality refers to the ability of users to control the extent to which personalization influences their recommendations. Users should be able to adjust the degree of algorithmic intervention in ranking models, ensuring that personalized content aligns with their preferences without fully overriding serendipitous discovery. In practice, this could be implemented via simple UI controls like sliders: for instance, a user might reduce the proportion of exploitative recommendations to discover new content. Prior research has explored interactive mechanisms to let users dial such trade-offs [30, 53]. For example, allowing users to manipulate a *personalization–diversity slider* was found to significantly affect their satisfaction and perceived recommendation quality [54]. In a field experiment with an interactive movie RS, users with different exploration preferences were given interfaces to control how broad or narrow their recommendations were; interestingly, the level of exploration control a user chose was strongly predictive of their eventual satisfaction with the system [54]. Moreover, a longitudinal study on music recommendations showed that when users are given a personalization slider for their playlist, their chosen personalization level can evolve over time: an initially high personalization setting might be gradually reduced as the user seeks more variety [36]. This suggests that users value being able to dynamically re-balance personalization as their needs change. Future RSs could allow users to maintain multiple recommendation profiles, dynamically re-weighting their preferred level of personalization across different contexts (e.g., a highly personalized profile for news and a less exploitative profile for exploratory browsing).

Granularity concerns the level of detail at which users can modify RS parameters or signals. Fine-grained control means users can adjust individual system features at different levels of depth, and immediately see how these adjustments impact their recommendations. Instead of treating the RS as a black box, platforms could provide controllable dials for key factors, e.g., a user could increase the weight of *topic relevance* while decreasing that of *recently viewed* items. Users should be able to inspect which features (e.g., engagement metrics such as number of likes, content attributes like genre or source) are driving the recommendations and then modify them. The idea of actively shaping the outcome of RSs in this way resonates with users' intentions [33, 52]. Prior studies have accordingly investigated various interfaces for feature-level control. For example, TasteWeights [7] allowed users to visually adjust the weights of different influencing factors in a hybrid music recommender. In the movie domain, a plug-in system called MovieBrain added interactive settings like genre-based filters and personalized hybrid weighing, which improved users' perception of the recommendation quality [15]. Relatedly, researchers have developed UIs for interactive feature weighing where users can tune the importance of content attributes and immediately see the adjusted recommendation list [37, 41, 45, 53]. Recent work on multi-objective RSs also shows that letting users adjust the importance of objectives (e.g., relevance vs. novelty) can improve their satisfaction and understanding of the system [12, 14]. In essence, granularity turns RS control into a two-way conversation: the system offers suggestions while the user can fine-tune the dials (feature weights, filters, etc.) to clarify their intent. Such transparency and contestability at the feature level do not only improve the explainability of results but also ensure that users with diverse needs feel empowered by the RS rather than constrained by it [52]. While major platforms have started providing more granular controls thanks to regulatory incentives, their efforts remain quite limited [31].

Directing content curation. This category focuses on the *scope* and *quality* of user intervention, determining *what* aspects of content the user can influence.

Diversity refers to mechanisms that enable users to explore various perspectives in their feed, reducing echo chamber effects while maintaining relevance. To come out of echo chambers, users should be able to adjust how diverse or novel their recommendation lists are. One approach is to provide a *diversity slider*, coupled with the personalization control mentioned above, so that users can explicitly balance highly focused recommendations with more eclectic ones. Studies have found that users who had greater access to diversity settings showed higher satisfaction when those controls aligned with their personal exploration preference [36, 54]. Moreover, allowing users to create multiple preference profiles (for different genres or contexts) has been suggested as a way to increase diversity [50]: e.g., a user could maintain one profile for news content and another for entertainment, effectively preventing any single profile from over-personalizing and thereby preserving a wider content spectrum. However, users' perception of diversity can be tricky, as highlighted by recent research that revealed that users tend to favor intuitive diversity metrics (such as diversity in genres or source types) over abstract algorithmic diversity measures like dissimilarity-optimized ranking [13]. Additionally, users quickly reach a point of saturation: beyond a certain level, small increases in objective diversity are not even noticed [13]. This aligns with observations that users often perceive diversity in non-linear ways; a moderate jump from very narrow to somewhat varied content yields a big increase in perceived diversity, but further broadening beyond the user's interest zone may decrease returns or even confuse the user [13]. Furthermore, an important paradox emerged in a study of multi-objective recommendations: although many users claim they want novel and diverse items, their actual interactions often gravitate toward the familiar [14]. This suggests that simply injecting diversity is not enough; the system should also help users make sense of novel content. Effective design could include showing explanatory cues or grouping diverse items with brief descriptions, so that users understand why those less familiar recommendations appear [14]. By introducing user-driven diversity controls, alongside supportive explanations, RSs can enable exploration while maintaining a sense of relevance, allowing users to deliberately broaden their horizons without feeling lost or mis-targeted.

Authoritativeness concerns emphasizing *trustworthy*, reliable sources in recommendations, which is a crucial aspect in domains like news, health, or any context where disinformation is a risk. This perspective gives users (and, by extension, the system) the ability to prioritize content based on credibility signals. RSs can integrate these preferences by incorporating source credibility metrics into the ranking process. For example, a platform might allow users to activate a "credibility boost" functionality which, when enabled, ranks items from reputable outlets higher in their feed. Internally, this could be implemented by increasing the weight of a source quality feature in the ranking algorithm. Recent research prototypes have demonstrated how multi-objective ranking can incorporate ethical or quality-oriented constraints without severely degrading user experience by treating the ranking task as a constrained optimization problem, where the maximization of predicted user engagement is subject to constraints on diversity, fairness, or credibility of content [60]. While such optimization can be computationally heavy, new methods have been proposed to predict the optimal trade-offs much faster, enabling these credibility and quality considerations to be applied in real time [60]. From a user's perspective, authoritativeness controls might also be presented in more transparent ways: for example, showing trust labels or source ratings alongside content, and giving users the option to filter or sort content by these indicators. User studies indicate that when users are aware of the trustworthiness of content (through labels or explanations), they often appreciate the ability to favor trustworthy sources, especially in high-stakes contexts like health information retrieval [60]. Enabling users to incorporate authoritativeness signals into RSs and clearly explaining them aligns with the requirements of Measures 22.2 and 22.3 of the CCD, as outlined in Section 2. Ultimately, by allowing users to easily elevate authoritative content (or demote dubious content), future RSs can serve both individuals' priorities and societal interest, fostering an ecosystem where quality information is more readily discovered.

Proposing a compliance standard. The legal analysis in Section 2 outlined how user empowerment through explanations and effective control, in DSA and CCD, and diversity and authoritativeness in content curation and moderation, in CCD and EMFA, are fundamental principles of the emerging EU regulatory framework

around RSs. Such principles have been confirmed also by the analysis of VLOPs' reports in Section 3, which, despite highlighting diverging interpretations, underscored the need for substantive user control options [35] and the promotion of trustworthy and diverse content as a means to combat disinformation [28]. These principles are aimed at realizing meaningful personalization, intended as the conjunction of users' awareness and active involvement in informing recommendations.

The four design perspectives introduced in this section can represent a framework for platforms' compliance with the EU legal requirements on RSs: moving from opaque, one-size-fits-all algorithms to transparent, user-steerable ecosystems where people can co-curate the content they consume. As mentioned above, one of the main problems regarding the evaluation of how transparent and controllable recommendations should look like stems from the absence of guidelines on how to interpret and operationalize the related regulatory requirements. The design framework we propose would allow users and regulators to evaluate the transparency and controllability of RSs according to proportionality and granularity, from the perspective of algorithmic choice, and diversity and authoritativeness, from the perspective of content curation. Platforms' RSs could be ranked through a standardized score measuring the extent to which users can influence their functioning across these four perspectives, resulting in an easily accessible visualization that compares the different RSs used by each VLOP. To connect our design framework with an initial standard for platforms to implement DSA-oriented RSs, we propose four checkpoints corresponding to each aspect outlined in our design framework.

(1) **Proportionality**: *To what extent can users decide the proportional impact of different RS types, policies and data on the algorithmic logic?*
(2) **Granularity**: *How deep and specific can users' intervention on algorithmic parameters and weights be?*
(3) **Diversity**: *Which metrics are adopted to determine how diverse the recommended content is and to what extent can users intervene on them?*
(4) **Authoritativeness**: *Based on the platform's definition of credible and trustworthy sources, can users decide how frequently they want to interact with authoritative content?*

5 Conclusion

Understanding and controlling the recommendation process in social media is a complex endeavor, in which those who succeeded the most are now identified as VLOPs under the DSA. With its requirements on RSs, the DSA aims to empower users through transparency and control, by allowing them to influence the sociotechnical infrastructure by which they have been influenced for more than two decades. However, the way in which such requirements have been implemented until now does not trace a consistent and substantive realization of the principle of user empowerment. The audits of Instagram and TikTok are emblematic of how the lack of guidelines for the implementation of art. 27 and 38 DSA can lead to disparate outcomes regarding which information, design features and control options are considered compliant under the DSA. These considerations allow us to come back to our first research question.

Addressing RQ1. *We argue that there is currently no unified interpretation nor consistent application of what the DSA provisions on RSs entail for VLOPs and, ultimately, user experience. However, related regulations such as CCD and EMFA can help clarify the path forward in terms of compliance.*

The first problem in this regard is the lack of definition of "parameters", "criteria" and "options", leading to a lexical vacuum which is often filled by the benchmarking convenience of auditors and platforms. The second problem is the current absence of legal guidance on what meaningful user control would consist in within this regulatory context: again, while the definition of thresholds is left to VLOPs, the benchmarking process, shared with auditors, may eventually capture the development of standards that should pertain to regulators. While our analysis has been based on three VLOPs, its results can be strengthened by considering the documentation pertaining to other VLOPs and VLOSEs and including also the (currently scarce) evidence on the DSA enforcement process by the Commission. As a result of the wider picture traced by the intersections of the DSA with related EU digital regulations, there is the normative ground to advance a meaningful personalization of online experience, intended as the ability for users to choose not only which type of content they want to see but also how they want to be exposed to such content. Meaningful personalization can be substantiated by allowing users, on the one side, to intervene on the design and logic of RSs through proportional and granular control options, and, on the other side, to select the degree of authoritativeness of the information they engage with and the diversity of the modalities through which they encounter such information. These perspectives on how to shape user empowerment could be applied through an actionable framework for standardizing compliance.

Addressing RQ2. *Users can be empowered to achieve a meaningful personalization of their online experience through the operationalization of proportionality, granularity, diversity and authoritativeness. These perspectives contribute to a design framework for transparent and controllable RSs that may inform compliance standards for platforms under the DSA.*

The limitations of our contribution mainly concern its theoretical nature, which calls for empirical development and testing. In particular, further research should verify the effectiveness of our proposed approach through real-world experiments on user control of social media RSs, also by taking advantage of the forthcoming data access mechanism enabled by art. 40(4) DSA, which is now fully operational following the Commission's approval of the related delegated act [21]. Directions worth exploring include: the empirical evaluation of the impact of personalization controls on user engagement, satisfaction, and perceived agency; the implications of feature-level contestability for interface design and users' awareness and trust; how to support exploration without sacrificing relevance or introducing cognitive overload; how to integrate authoritativeness signals into ranking processes to improve informed user choice, particularly in sensitive domains like news or health. Research efforts should not only aim to evaluate compliance, but also to redefine user empowerment through a meaningful personalization approach that views RSs as controllable and transparent tools to improve user awareness and self-determination in (very large) online platforms.

References

[1] [n. d.]. Regulation (EU) 2022/2065 of the European Parliament and of the Council of 19 October 2022 on a Single Market For Digital Services and amending Directive 2000/31/EC (Digital Services Act). https://eur-lex.europa.eu/eli/reg/2022/2065/oj/eng

[2] [n. d.]. Regulation (EU) 2024/1083 of the European Parliament and of the Council of 11 April 2024 establishing a common framework for media services in the internal market and amending Directive 2010/13/EU (European Media Freedom Act). https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32024R1083

[3] Taiwo Amoo and Hershey Friedman. 2001. Do Numeric Values Influence Subjects' Responses to Rating Scales? *Journal of International Marketing and Marketing Research* 26 (01 2001), 41–46. https://ssrn.com/abstract=2423064

[4] James Auger. 2013. Speculative design: crafting the speculation. *Digital Creativity* 24, 1 (2013), 11–35.

[5] Eric PS Baumer, Mark Blythe, and Theresa Jean Tanenbaum. 2020. Evaluating design fiction: The right tool for the job. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference*. 1901–1913.

[6] Svetlin Bostandjiev, John O'Donovan, and Tobias Höllerer. 2012. TasteWeights: a visual interactive hybrid recommender system. In *Proceedings of the Sixth ACM Conference on Recommender Systems* (Dublin, Ireland) *(RecSys '12)*. Association for Computing Machinery, New York, NY, USA, 35–42. doi:10.1145/2365952.2365964

[7] Svetlin Bostandjiev, John O'Donovan, and Tobias Höllerer. 2012. TasteWeights: a visual interactive hybrid recommender system. In *Sixth ACM Conference on Recommender Systems, RecSys '12, Dublin, Ireland, September 9-13, 2012*, Padraig Cunningham, Neil J. Hurley, Ido Guy, and Sarabjot Singh Anand (Eds.). ACM, 35–42. doi:10.1145/2365952.2365964

[8] Federica Cena, Cristina Gena, Pierluigi Grillo, Tsvi Kuflik, Fabiana Vernero, and Alan J. Wecker. 2017. How scales influence user rating behaviour in recommender systems. *Behaviour & Information Technology* 36, 10 (2017), 985–1004. doi:10.1080/0144929X.2017.1322145

[9] European Commission. 2024. Commission opens formal proceedings against TikTok on election risks under the Digital Services Act. https://ec.europa.eu/commission/presscorner/detail/en/ip_24_6487

[10] European Commission. 2024. Commission opens formal proceedings against TikTok on election risks under the Digital Services Act. https://ec.europa.eu/commission/presscorner/detail/en/ip_24_6487

[11] European Commission. 2024. DSA.100121 - TikTok - INVESTIGATION INTO COMPLIANCE WITH ARTICLES 34, 35 AND 42(4). https://ec.europa.eu/newsroom/dae/redirection/document/108626

[12] Patrik Dokoupil, Ludovico Boratto, and Ladislav Peska. 2024. Comparing User Interfaces for Customizing Multi-Objective Recommender Systems. In *Proceedings of the 11th Joint Workshop on Interfaces and Human Decision Making for Recommender Systems* (Bari, Italy) *(IntRS@RecSys'24)*. CEUR-WS.org, Aachen, Germany, 15–28. https://ceur-ws.org/Vol-3815/paper2.pdf

[13] Patrik Dokoupil, Ludovico Boratto, and Ladislav Peska. 2024. User Perceptions of Diversity in Recommender Systems. In *Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization, UMAP 2024, Cagliari, Italy, July 1-4, 2024*. ACM, 212–222. doi:10.1145/3627043.3659555

[14] Patrik Dokoupil, Ladislav Peska, and Ludovico Boratto. 2023. Looks Can Be Deceiving: Linking User-Item Interactions and User's Propensity Towards Multi-Objective Recommendations. In *Proceedings of the 17th ACM Conference on Recommender Systems* (Singapore, Singapore) *(RecSys '23)*. Association for Computing Machinery, New York, NY, USA, 912–918. doi:10.1145/3604915.3608848

[15] Simon Dooms, Toon De Pessemier, and Luc Martens. 2014. Improving IMDb Movie Recommendations with Interactive Settings and Filters. In *Poster Proceedings of the 8th ACM Conference on Recommender Systems, RecSys 2014, Foster City, Silicon Valley, CA, USA, October 6-10, 2014 (CEUR Workshop Proceedings, Vol. 1247)*, Li Chen and Jalal Mahmud (Eds.). CEUR-WS.org. https://ceur-ws.org/Vol-1247/recsys14_poster19.pdf

[16] European Commission. 2023. Commission delegated regulation supplementing Regulation (EU) 2022/2065 of the European Parliament and of the Council, by laying down rules on the performance of audits for very large online platforms and very large online search engines. https://ec.europa.eu/newsroom/dae/redirection/document/99436

[17] European Commission. 2024. Commission opens formal proceedings against TikTok under the Digital Services Act. https://ec.europa.eu/commission/presscorner/detail/en/ip_24_926

[18] European Commission. 2024. Commission sends requests for information to YouTube, Snapchat, and TikTok on recommender systems under the Digital Services Act. https://digital-strategy.ec.europa.eu/en/news/commission-sends-requests-information-youtube-snapchat-and-tiktok-recommender-systems-under-digital

[19] European Commission. 2025. The Code of Conduct on Disinformation. https://digital-strategy.ec.europa.eu/en/library/code-conduct-disinformation

[20] European Commission. 2025. COMMISSION OPINION of 13.2.2025 on the assessment of the Code of Practice on Disinformation within the meaning of Article 45 of Regulation 2022/2065. https://ec.europa.eu/newsroom/dae/redirection/document/112679

[21] European Commission. 2025. Delegated act on data access under the Digital Services Act (DSA). https://digital-strategy.ec.europa.eu/en/library/delegated-act-data-access-under-digital-services-act-dsa

[22] EY. 2024. Independent Audit on Instagram. https://transparency.meta.com/sr/dsa-independent_audit_report_Aug_2023-Jun_2024_instagram

[23] EY. 2024. Independent Audit on YouTube. https://storage.googleapis.com/transparencyreport/report-downloads/dsa-audit-youtube_2023-8-28_2024-5-31_en_v1.pdf

[24] Matteo Fabbri. 2023. How and to which extent will the provisions of the Digital Services Act of the European Union impact on the relationship between users and platforms as information providers?. In *Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society*. 1002–1003.

[25] Matteo Fabbri. 2023. Self-determination through explanation: an ethical perspective on the implementation of the transparency requirements for recommender systems set by the Digital Services Act of the European Union. In *Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society*. 653–661.

[26] Matteo Fabbri. 2025. The Role of Requests for Information in Governing Digital Platforms Under the Digital Services Act: The Case of X. *Journalism and Media* 6, 1 (2025), 41.

[27] Google. 2024. Code of Practice of Disinformation – Report of Google for the period of 1 January 2024 to 30 June 2024. https://disinfocode.eu/reports-archive/reports-september-2024/

[28] Google. 2024. Report of Systemic Risk Assessments. https://storage.googleapis.com/transparencyreport/report-downloads/dsa-risk-assessment_2023-8-28_2023-8-28_en_v1.pdf

[29] Google Ireland Limited. 2024. DSA Audit Implementation Report. https://storage.googleapis.com/transparencyreport/report-downloads/dsa-audit-google-implementation_2023-8-28_2024-5-31_en_v1.pdf

[30] F. Maxwell Harper, Funing Xu, Harmanpreet Kaur, Kyle Condiff, Shuo Chang, and Loren Terveen. 2015. Putting Users in Control of their Recommendations. In *Proceedings of the 9th ACM Conference on Recommender Systems* (Vienna, Austria) *(RecSys '15)*. Association for Computing Machinery, New York, NY, USA, 3–10. doi:10.1145/2792838.2800179

[31] YouTube Help. 2025. Manage your recommendations & search results. https://support.google.com/youtube/answer/6342839?hl=en. [Accessed 13-07-2025].

[32] Yucheng Jin, Bruno De Lemos Ribeiro Pinto Cardoso, and Katrien Verbert. 2017. How do different levels of user control affect cognitive load and acceptance of recommendations?. In *IntRS@ RecSys*. 35–42.

[33] Daniel N Kluttz and Deirdre K Mulligan. 2019. Automated decision support technologies and the legal profession. *Berkeley Technology Law Journal* 34, 3 (2019), 853–890.

[34] Bart P. Knijnenburg, Svetlin Bostandjiev, John O'Donovan, and Alfred Kobsa. 2012. Inspectability and control in social recommenders. In *Proceedings of the Sixth ACM Conference on Recommender Systems* (Dublin, Ireland) *(RecSys '12)*. Association for Computing Machinery, New York, NY, USA, 43–50. doi:10.1145/2365952.2365966

[35] KPMG. 2024. DSA Assurance Report. https://sf16-va.tiktokcdn.com/obj/eden-va2/zayvwlY_fjulyhwzuhy[/ljhwZthlaukjlkulzlp/DSA_H2_2024/TikTok-DSA-Audit-Assurance-Report-Sep-2024.pdf

[36] Yu Liang and Martijn C. Willemsen. 2022. Exploring the longitudinal effects of nudging on users' music genre exploration behavior and listening preferences. In *RecSys '22: Sixteenth ACM Conference on Recommender Systems, Seattle, WA, USA, September 18 - 23, 2022*, Jennifer Golbeck, F. Maxwell Harper, Vanessa Murdock, Michael D. Ekstrand, Bracha Shapira, Justin Basilico, Keld T. Lundgaard, and Even Oldridge (Eds.). ACM, 3–13. doi:10.1145/3523227.3546772

[37] Yu Liang and Martijn C. Willemsen. 2022. Exploring the longitudinal effects of nudging on users' music genre exploration behavior and listening preferences. In *Proceedings of the 16th ACM Conference on Recommender Systems* (Seattle, WA, USA) *(RecSys '22)*. Association for Computing Machinery, New York, NY, USA, 3–13. doi:10.1145/3523227.3546772

[38] Ben Light, Jean Burgess, and Stefanie Duguay. 2018. The walkthrough method: An approach to the study of apps. *New media & society* 20, 3 (2018), 881–900.

[39] Laura Marciano, Bernadka Dubicka, Lucía Magis-Weinberg, Rosalba Morese, Kasisomayajula Viswanath, and René Weber. 2025. Digital Media, Cognition, and Brain Development in Adolescence. In *Handbook of Children and Screens*. Springer, Cham, 21–29.

[40] Silvia Milano, Mariarosaria Taddeo, and Luciano Floridi. 2020. Recommender systems and their ethical challenges. *Ai & Society* 35 (2020), 957–967.

[41] Martijn Millecamp, Nyi Nyi Htun, Yucheng Jin, and Katrien Verbert. 2018. Controlling Spotify Recommendations: Effects of Personal Characteristics on Music Recommender User Interfaces. In *Proceedings of the 26th Conference on User Modeling, Adaptation and Personalization* (Singapore, Singapore) *(UMAP '18)*. Association for Computing Machinery, New York, NY, USA, 101–109. doi:10.1145/3209219.3209223

[42] Christian Montag, Bernd Lachmann, Marc Herrlich, and Katharina Zweig. 2019. Addictive features of social media/messenger platforms and freemium games against the background of psychological and economic theories. *International journal of environmental research and public health* 16, 14 (2019), 2612.

[43] John O'Donovan, Barry Smyth, Brynjar Gretarsson, Svetlin Bostandjiev, and Tobias Höllerer. 2008. PeerChooser: visual interactive recommendation. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Florence, Italy) *(CHI '08)*. Association for Computing Machinery, New York, NY, USA, 1085–1088. doi:10.1145/1357054.1357222

[44] Denis Parra and Peter Brusilovsky. 2015. User-controllable personalization: A case study with SetFusion. *International Journal of Human-Computer Studies* 78 (2015), 43–67. doi:10.1016/j.ijhcs.2015.01.007

[45] Denis Parra, Peter Brusilovsky, and Christoph Trattner. 2014. See what you want to see: visual user-driven approach for hybrid recommendation. In *19th International Conference on Intelligent User Interfaces, IUI 2014, Haifa, Israel, February 24-27, 2014*, Tsvi Kuflik, Oliviero Stock, Joyce Yue Chai, and Antonio Krüger (Eds.). ACM, 235–240. doi:10.1145/2557500.2557542

[46] Meta Platforms. 2024. Audit Implementation Report for Instagram Regarding Regulation (EU) 2022/2065, the Digital Services Act (DSA). https://transparency.meta.com/sr/dsa-audit_implementation_report_Sept_2024_instagram

[47] Meta Platforms. 2024. Our approach to explaining ranking. https://transparency.meta.com/features/explaining-ranking/

[48] Meta Platforms. 2024. Regulation (EU) 2022/2065 Digital Services Act (DSA) - Systemic Risk Assessment and Mitigation Report for Instagram. https://scontent-dus1-1.xx.fbcdn.net/v/t39.8562-6/468481711_1994741217693263_5869766641031282981_n.pdf?_nc_cat=106&ccb=1-7&_nc_sid=b8d81d&_nc_ohc=A0tw9NWkjwkQ7kNvgGkhfWJ&_nc_zt=14&_nc_ht=scontent-dus1-1.xx&_nc_gid=ADiAYO2x9S8umqamoVAcgU-&oh=00_AYDsWtjJiy1_8koPrkj8eywHJVTPGd3VMWNNdr-GZpMdwg&oe=67970FF5

[49] Yao Qin, Bahiyah Omar, and Alessandro Musetti. 2022. The addiction behavior of short-form video app TikTok: The information quality and system quality perspective. *Frontiers in Psychology* 13 (2022), 932805.

[50] Urbano Reviglio and Matteo Fabbri. 2024. Navigating the Digital Services Act: Scenarios of Transparency and User Control in VLOPs' Recommender Systems. *NORMalize 2024: The Second Workshop on the Normative Design and Evaluation of Recommender Systems, October 18, 2024, co-located with the ACM Conference on Recommender Systems 2024 (RecSys 2024), Bari, Italy* (2024).

[51] Arianna Sala, Lorenzo Porcaro, and Emilia Gómez. 2024. Social Media Use and adolescents' mental health and well-being: An umbrella review. *Computers in Human Behavior Reports* 14 (2024), 100404.

[52] Christopher Starke, Ljubiša Metikoš, Natali Helberger, and Claes de Vreese. 2025. Contesting personalized recommender systems: a cross-country analysis of user preferences. *Information, Communication & Society* 28, 1 (2025), 41–60. doi:10.1080/1369118X.2024.2363926

[53] Ruixuan Sun, Avinash Akella, Ruoyan Kong, Moyan Zhou, and Joseph A. Konstan. 2023. Interactive Content Diversity and User Exploration in Online Movie Recommenders: A Field Experiment. *International Journal of Human–Computer Interaction* 0, 0 (2023), 1–15. doi:10.1080/10447318.2023.2262796 arXiv:https://doi.org/10.1080/10447318.2023.2262796

[54] Ruixuan Sun, Avinash Akella, Ruoyan Kong, Moyan Zhou, and Joseph A. Konstan. 2024. Interactive Content Diversity and User Exploration in Online Movie Recommenders: A Field Experiment. *Int. J. Hum. Comput. Interact.* 40, 22 (2024), 7233–7247. doi:10.1080/10447318.2023.2262796

[55] Petros Terzis, Michael Veale, and Noëlle Gaumann. 2024. Law and the Emerging Political Economy of Algorithmic Audits. In *The 2024 ACM Conference on Fairness, Accountability, and Transparency*. 1255–1267.

[56] TikTok. [n. d.]. How TikTok recommends content. https://support.tiktok.com/en/using-tiktok/exploring-videos/how-tiktok-recommends-content

[57] TikTok. 2024. DSA Audit Implementation Report. https://sf16-va.tiktokcdn.com/obj/eden-va2/zayvwlY_fjulyhwzuhy[/ljhwZthlaukjlkulzlp/DSA_H2_2024/TikTok-DSA-Audit-Implementation-Report-Sep-2024.pdf

[58] TikTok. 2024. DSA RISK ASSESSMENT REPORT 2023. https://sf16-va.tiktokcdn.com/obj/eden-va2/zayvwlY_fjulyhwzuhy[/ljhwZthlaukjlkulzlp/DSA_H2_2024/TikTok-DSA-Risk-Assessment-Report-2023.pdf

[59] TikTok. 2024. TikTok's STEM feed now available to all users by default. https://newsroom.tiktok.com/en-gb/tiktok-stem-feed-now-available-to-all-users-by-default

[60] Yegor Tkachenko, Wassim Dhaouadi, and Kamel Jedidi. 2022. Scaling up Ranking under Constraints for Live Recommendations by Replacing Optimization with Prediction. *CoRR* abs/2202.07088 (2022). arXiv:2202.07088 https://arxiv.org/abs/2202.07088

[61] Cindy S Vincent and Adam Gismondi. 2021. Fake news, reality apathy, and the erosion of trust and authenticity in American politics. In *Democracy in the Disinformation Age*. Routledge, 79–98.

[62] YouTube Help. 2024. Advanced search (Filters). https://support.google.com/youtube/answer/111997?hl=en&ref_topic=9257501&sjid=6235270588440401107-EU

[63] YouTube Help. 2024. Manage your recommendations and search results. https://support.google.com/youtube/answer/6342839?hl=en&ref_topic=9257501&sjid=6235270588440401107-EU

[64] Monika Zalnieriute. 2021. " Transparency Washing" in the Digital Age: A Corporate Agenda of Procedural Fetishism. *Critical Analysis L.* 8 (2021), 139.